



Image Analysis Collaboratory



ACITE

Core for Imaging Technology & Education

The detectors used in fluorescence microscopy are **monochromatic**.

Microscope cameras are **not able to distinguish between different wavelengths** (they just collect photons), you need **fluorescence filters** to separate your fluorophores.

The detector converts photons in digital numbers (linear relation).

Detectors in Fluorescence Microscopy

The range of possible digital values is defined by the bit depth.

Each pixel in the digital image has one digital value that depends on the intensity of the signal emitted by the sample.





136	106	152	179
109	209	236	179
103	179	189	132
123	186	192	169

Digital Values = Pixel Intensity Value

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Fluorescence Microscopy

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Bit Depth

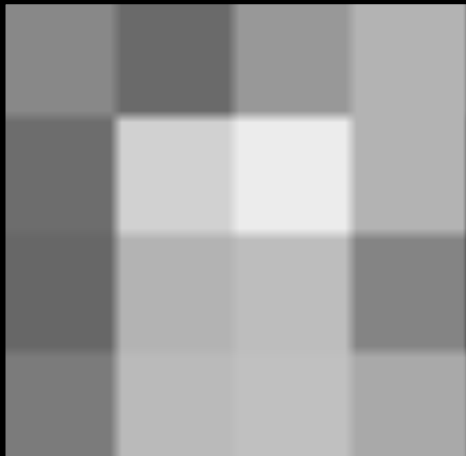
The bit depth is expressed in grey values.

The bit depth defines the range of possible digital values that

each pixel and have, usually 8, 12 or 16 bit.

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